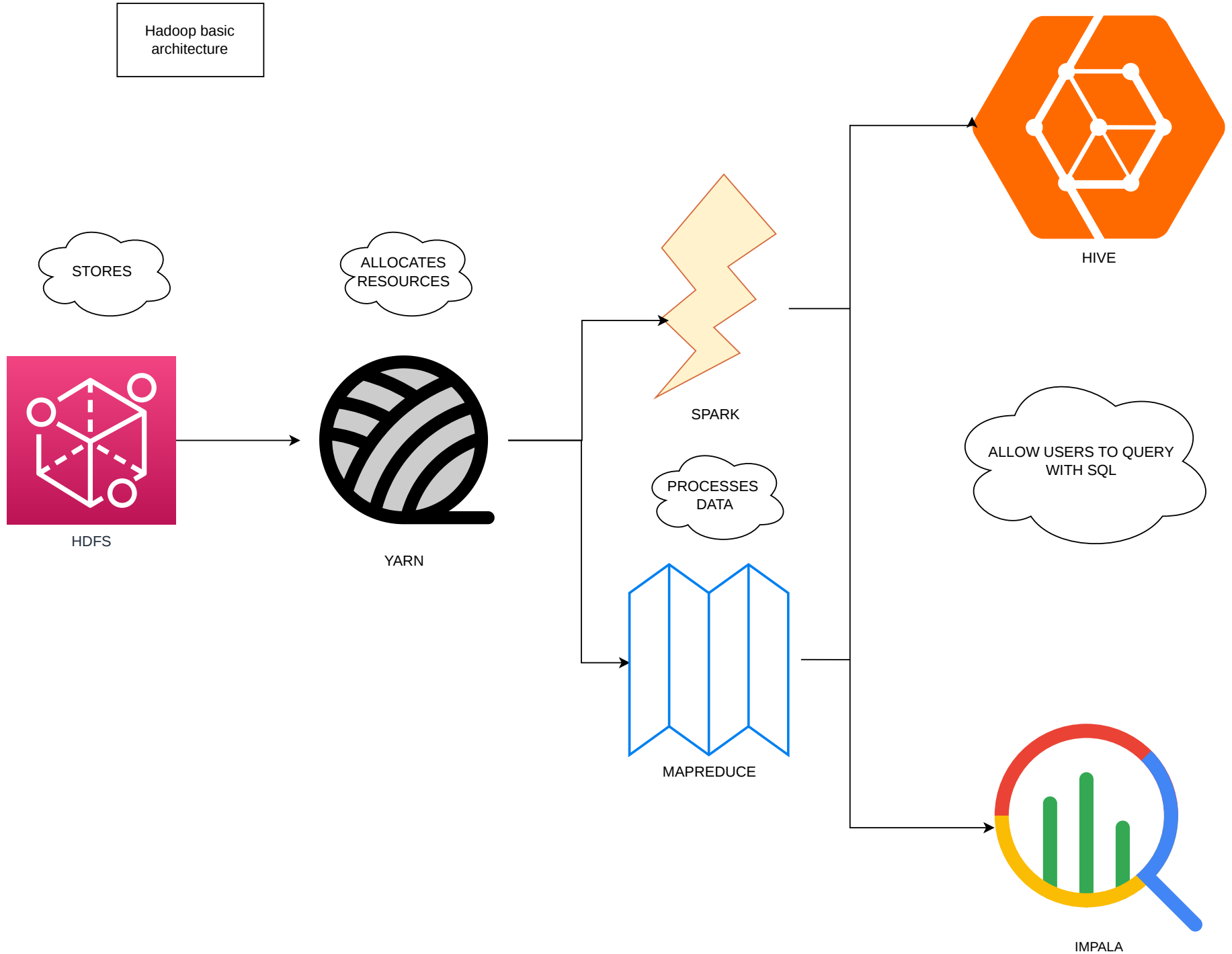
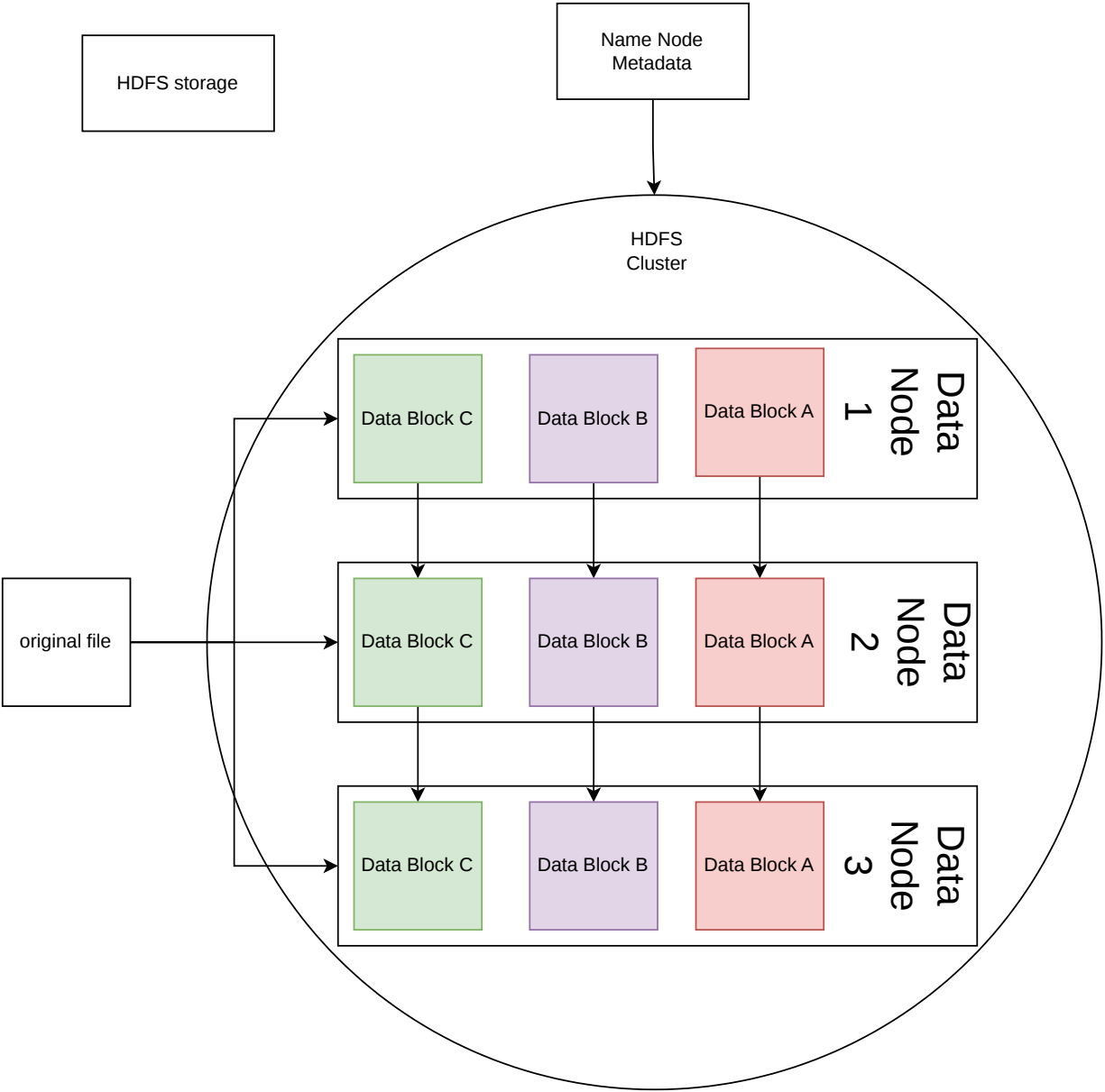


Hadoop basic architecture





YARN



spark app,
mapreduce job,
hive query etc

APPLICATION



manages
whole cluster

RESOURCE
MANAGER

ALLOCATES WORKER NODES

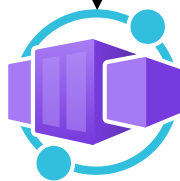


manages a
single node
each

NODE
MANAGERS

1. ASKS NODE MANAGER TO
START APPLICATION MASTER

REQUESTS RESOURCES FROM
RESOURCE MANAGER



does tasks

CONTAINER

CHECKS STATUS



manages
applications
across
nodes

APPLICATION
MASTER

MAPREDUCE

Input in HDFS → Input Splits → Map → Shuffle and Sort → Reduce → output saved to HDFS

SPARK



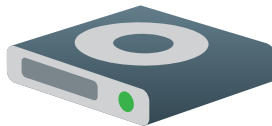
RESOURCE
MANAGER

Allocates
containers



NODE
MANAGERs

Launch
containers



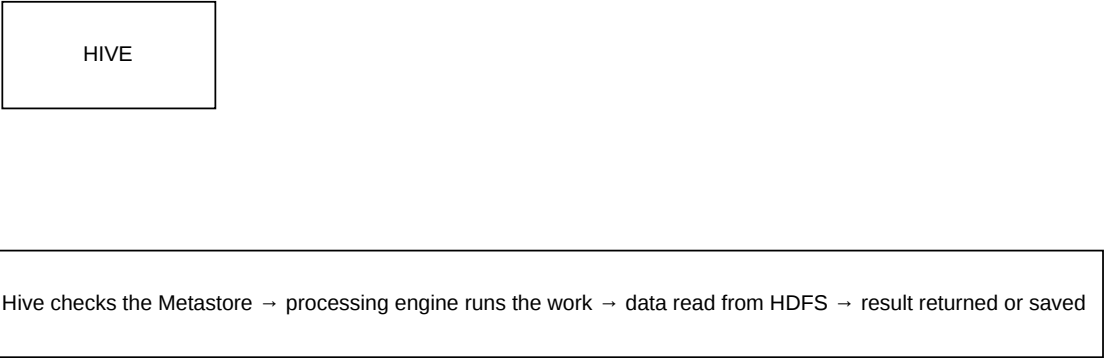
Driver

creates
stages / tasks



Executor

Runs tasks



HIVE

HiveQL query → Hive checks the Metastore → processing engine runs the work → data read from HDFS → result returned or saved